Date: 2026-05-01
Request ID: 100048474

Hotplate		
Element(s)	Al, Co, Pt, Si, Ti	
SOP#		
Cocktail	2 mL Di H2O, 10 mL conc. HCl, 10 mL conc. HNO3	
sample(s)	**weight (g)**	**volume (mL)**
200127713_1	0.2006	25
200127713_2	0.2006	25
200127714_1	0.2014	25
200127714_2	0.2014	25

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200127713_1	1/1	0.571	71.16	0.01
200127713_2	1/1	0.571	71.16	0.01

al result: 71.16 ppm
al result: 0.01 %
al oxide factor:
al oxide result: 0.00 %
al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm	%Co
200127713_1	1/1	0.170	21.19	0.00
200127713_2	1/1	0.170	21.19	0.00

co result: 21.19 ppm
co result: 0.00 %
co oxide factor:
co oxide result: 0.00 %
co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm	%Pt
200127713_1	1/1	0.051	6.36	0.00
200127713_2	1/1	0.051	6.36	0.00

pt result: 6.36 ppm
pt result: 0.00 %
pt oxide factor:
pt oxide result: 0.00 %
pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm	%Si
200127713_1	1/1	0.694	86.49	0.01
200127713_2	1/1	0.694	86.49	0.01

si result: 86.49 ppm
si result: 0.01 %
si oxide factor:
si oxide result: 0.00 %
si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm	%Ti
200127713_1	1/1	0.031	3.86	0.00
200127713_2	1/1	0.031	3.86	0.00

ti result: 3.86 ppm

ti result: 0.00 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *Not enough sample for duplicates.*

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200127714_1	1/1	0.418	51.89	0.01
200127714_2	1/1	0.418	51.89	0.01

al result: 51.89 ppm
al result: 0.01 %
al oxide factor:
al oxide result: 0.00 %
al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm	%Co
200127714_1	1/1	0.062	7.70	0.00
200127714_2	1/1	0.062	7.70	0.00

co result: 7.70 ppm
co result: 0.00 %
co oxide factor:
co oxide result: 0.00 %
co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm	%Pt
200127714_1	1/1	0.065	8.07	0.00
200127714_2	1/1	0.065	8.07	0.00

pt result: 8.07 ppm
pt result: 0.00 %
pt oxide factor:
pt oxide result: 0.00 %
pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm	%Si
200127714_1	1/1	0.468	58.09	0.01
200127714_2	1/1	0.468	58.09	0.01

si result: 58.09 ppm
si result: 0.01 %
si oxide factor:
si oxide result: 0.00 %
si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm	%Ti
200127714_1	1/1	0.005	0.62	0.00
200127714_2	1/1	0.005	0.62	0.00

ti result: 0.62 ppm
ti result: 0.00 %
ti oxide factor:
ti oxide result: 0.00 %
ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *Not enough sample for duplicates.*

A = crucible
B = crucible + sample
C = crucible + sample after drying

%LOI = ([B-A]-[C-A])/(B-A)*100%

ppm M+ = [conc.][volume][dilution]/[weight]

%M+ = [conc.][volume][dilution]/([weight]*10,000)

%MO = [oxide factor]*%M+

ppm M+ [dried] = ppm M+/([1-(%LOI)/100])

%M+ [dried] = %M+/([1-(%LOI)/100])

%MO [dried] = %MO/([1-(%LOI)/100])

%Cr(VI) = (1.733[mL FAS][N FAS])/([weight])

%Cr2O3 = 1.462*[%Cr(VI)]

%Cr(III) = total_Cr - Cr(VI)